



HOPE FOR THE BEST &
PREPARE FOR THE WORST .

SLUM REHABILITATION

DESIGNING LIKE PEOPLE MATTERS

MENTOR

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PROJECT BY

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‘HUM’ is the word used for a democratic society, where’s there is no differentiation between rich & poor, high & low and inequalities. Nonetheless, its people who change the world.

‘By the people, for the people, from the people’

This hypothesis is taken and transformed into this phenomenon involving this essence and inherent features like a small interactive spaces for people to come together and a community centre, etc.

INTRODUCTION/ BACKGROUND

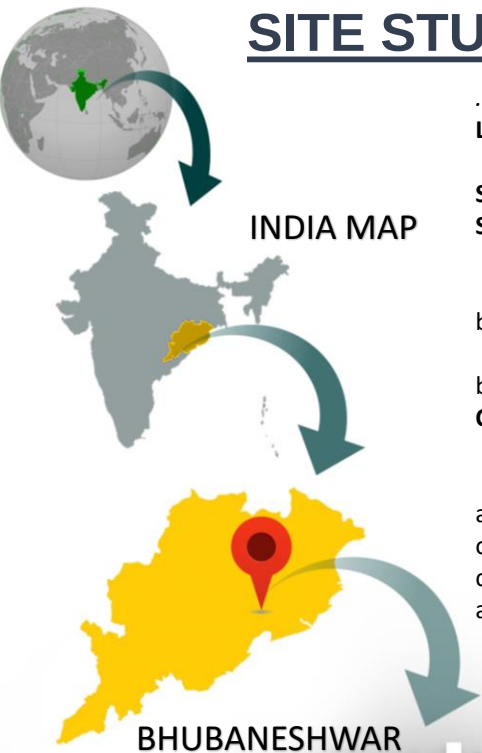
- The objective is to design sustainable buildings and settlements, which are resilient and can cope with natural and anthropogenic disasters.
- The main focus is on marginalized sections of the society.
- We introduce **HUM** as our theoretical concept of 'Sense of belonging' conveying no differences between any individuals.

WHY DO WE NEED DISASTER RISK REDUCTION RESILIENT ARCHITECTURE :

- We cannot stop natural disasters but we can arm ourselves with knowledge , so many lives wouldn't have to be lost if there is enough disaster preparedness.
 - And the most ratio of preparedness can happen with the building forms, orientation and function.
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- In this case , the site is faced by natural disasters like cyclone , flood , earthquake.
Therefore ,
 1. In case of cyclone- we can reduce the flow of wind by changing the orientation of the building
 2. In case of earthquake – running bands throughout the building and providing shear walls can reduce the damage of the structure.



SITE STUDY AND ANALYSIS



INDIA MAP

BHUBANESHWAR

LOCATION : Mouza bhoynagar, Bayababa slum in Ward no.35, Bhubaneswar

SITE TOPOGRAPHY : Flat land

SOIL TYPE : Top soil- A mix soil (deposits, stone, stone, sand)
Hard Laterite rock - 1m height below
Ground water table - 6 m height below

CLIMATE : Tropical Savanna Climate (HOT AND HUMID)

In Bhubaneswar, the wet season is oppressive and overcast, the dry season is muggy and mostly clear, and it is hot year round.



SWOT ANALYSIS:

Strength-

The site is located at the center of the city.
The site has good road network and public transport connectivity.

Weakness-

Canal running along the south-west side of the site. This unhygienic canal may spread diseases like dengue, etc.

Opportunities-

As site is in the centre of city, it is surrounded by malls, hospitals, parks and educational institutions providing the availability of jobs for the users.

Threats-

Disasters like cyclone and flood are threat to this site.
Densely populated area hence high risk during this ongoing pandemic crisis.

1. BHUBANESHWAR RAILWAY STATION



2. PETROL PUMP



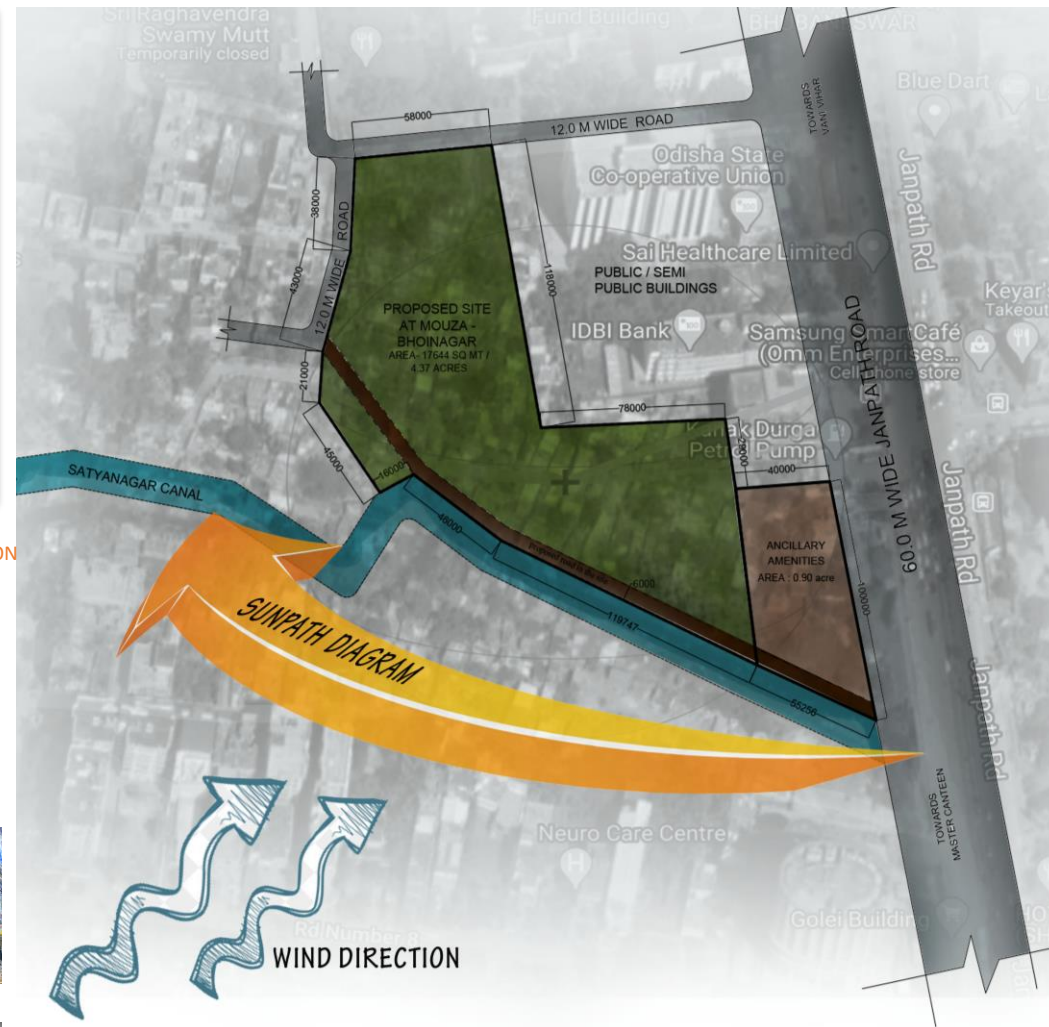
3. BMC MALL



4. BIJU PATNAIK AIRPORT



5. KALI TEMPLE



LEARNINGS FROM CASE STUDIES

LOCATION :Madhusudanpalli,Rourkela ,
Orissa.

- Population – 2,73,217 (2011).
- Total Slum Population – 1,32,000(2011)
- Total Area – 31.6 Sq.Km

Scheme : RAJIV AWAS YOJNA

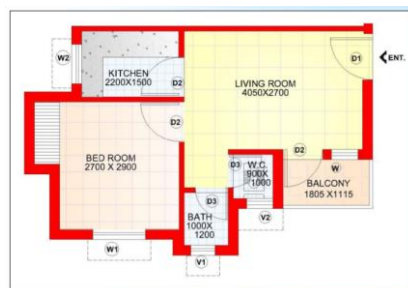
PROVIDING BASIC PHYSICAL INFRASTRUCTURE :

- 1.Water supply
- 2.Solid waste management
- 3.Sanitation
- 4.Roads
- 5.Rain water harvesting
- 6.Street lighting
- 7.Drainage.

SITE PLAN



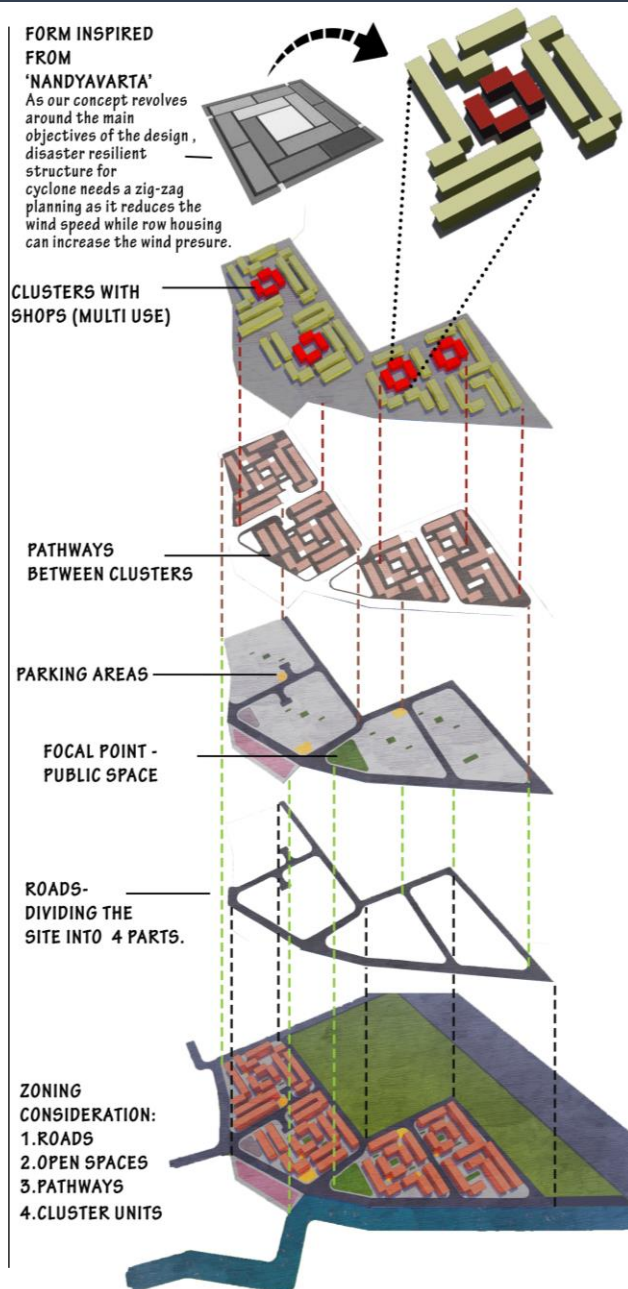
UNIT PLAN



DESIGN CONCEPT

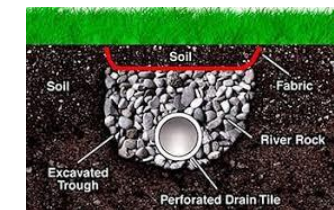


- The concept is basically divided into three main aspects:
- 1. DISASTER RESILIENT ARCHITECTURE:**
The intentional design of buildings,landscapes, communities are reasons in response to vulnerabilities to disaster and disruption of normal life.
 - 2. SUSTAINABILITY: (SELF-DEPENDENT)**
 - Energy efficient- solar panels.
 - Water efficient – rain water harvesting , waste water management
 - Solid waste management – Bio gas plant
 - 3. BARRIER FREE DESIGN :**
“FOCAL POINT-
We chose the centre of the site along the proposed road as a prominent spot for a socio-public place. A park which is designed according to the considerations of barrier free planning.



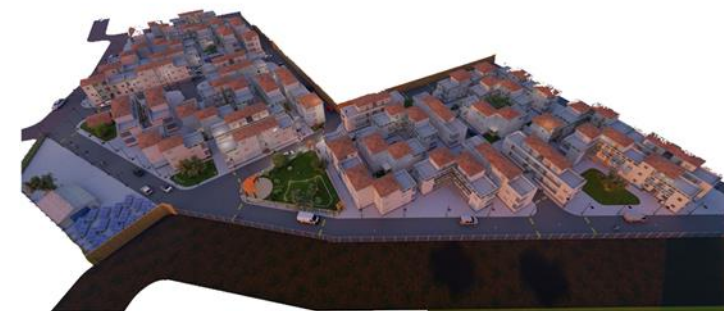
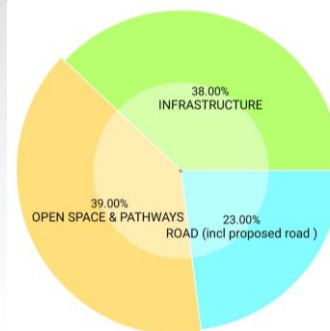
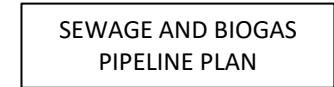
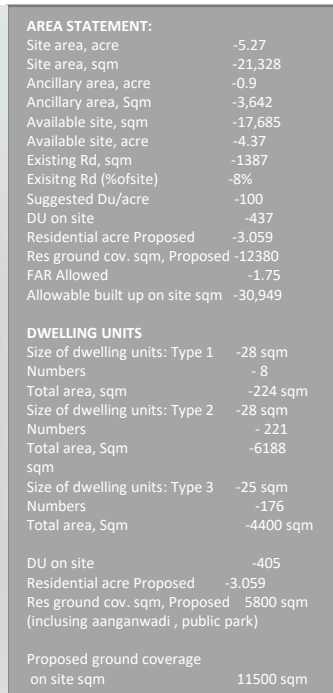
DESIGN FEATURES:

- Fire safety.
- Green pockets
- Shops with interactive space
- French drain
- Focal point of the site- centre park
- Porous road
- Waste water management (osho garden)
- Bio-gas plant
- Solar panel (application – solar street lights)



DESIGN DETAILS – I

(will include Drawings, 3D views, sketches, etc.)



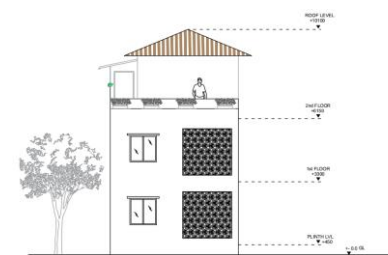
DESIGN DETAILS – II (Plans , sections, elevations.)



ELEVATION AT A



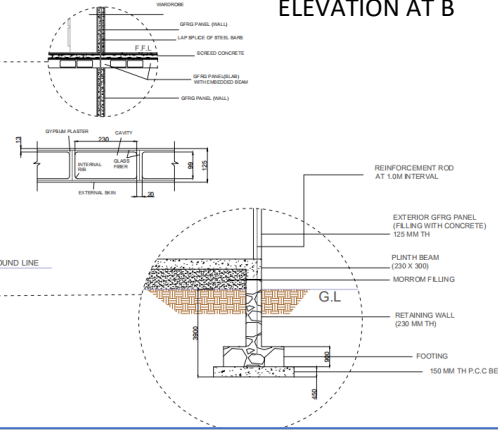
ELEVATION AT B



ELEVATION AT C



SECTION A-A'



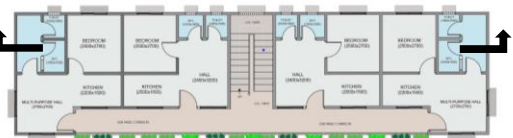
SECTION C-C'



UNIT PLAN (3D VIEW)



2ND FLOOR



1ST FLOOR



GROUND FLOOR

TYPE 1 (47%)

TYPE 2 (42%)

TYPE 3 (11%)



GROUND FLOOR



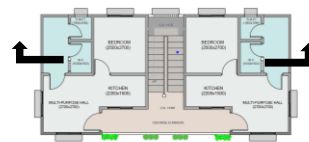
1st FLOOR



2nd FLOOR



GROUND FLOOR



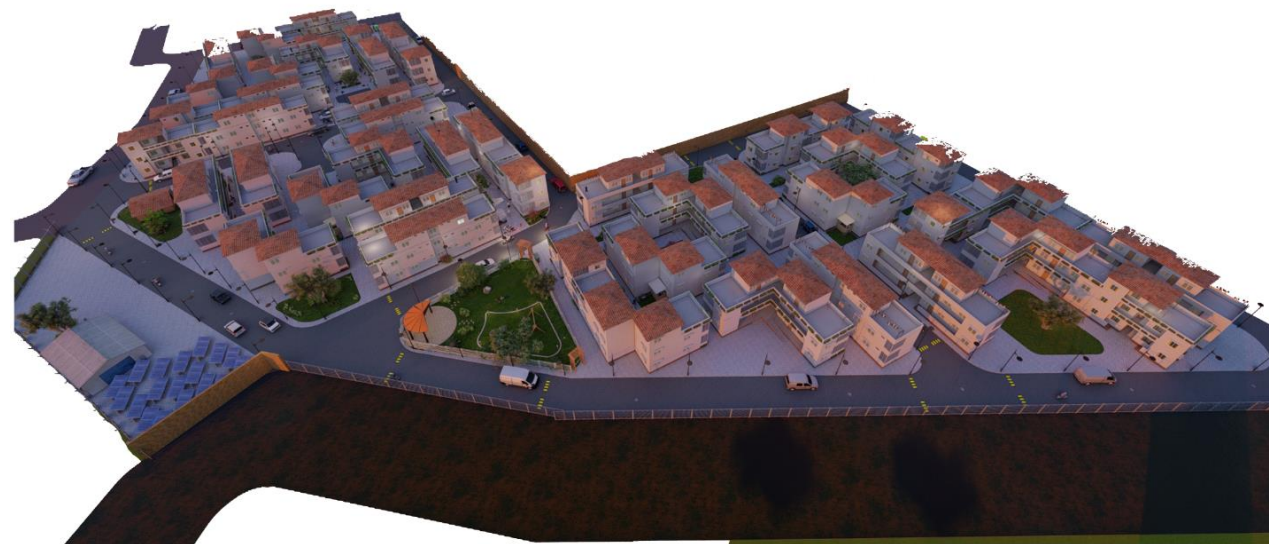
1ST FLOOR



2nd FLOOR

DESIGN DETAILS – III

(will include Drawings, 3D views, sketches, etc.)



DISASTER RESILIENCE STRATEGIES

CYCLONE

AT SITE LEVEL

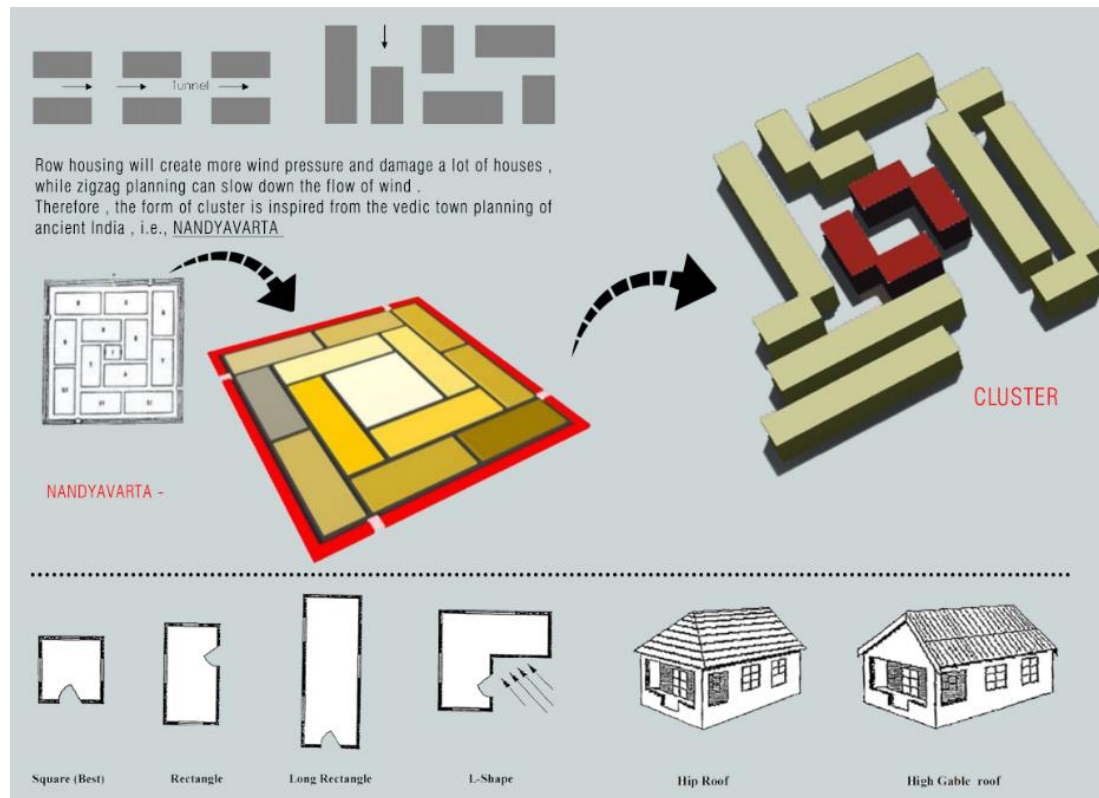
According to the Disaster Resilient structure (design objective), one of the main cause of the building destruction due to cyclone, floods in this case, is the orientation of the building block.

In case of construction of group of buildings, a cluster arrangement, row housing should be avoided as it creates more wind pressure causing lot of damage.

UNIT LEVEL

Shape is the most important single factor in determining the performance of buildings in cyclones.

The square plan is better than the rectangle since it allows high winds to go around them.



FLOOD

For Flood resilient strategies applied in our design is

1. Plinth level kept high
2. French drain – it allows water to drain naturally from your yard, so that water cannot enter into the foundation.
3. Material of the wall – GFRG panel Which is fire resistance upto 1000 degree

FIRE HAZARD

- For fire safety, periphery road of 5.0 m wide is running along the site.
- Cul-de-sacs are given, so that the fire brigade can reach the internal houses of each of the 4 parts which is divided by roads.
- Every house is at a minimum radius of 20.0m to the fire brigade road so that, even if engine may not be able to go inside, so the pipes can.

EARTHQUAKE

In case of earthquake,

1. Material of the wall – GFRG panel, which is earthquake resistant.
2. When the hollow blocks of the GFRG panels are filled with concrete and reinforcement, it becomes as much strong as shear wall.

MATERIALS USED

LATERITE BLOCK

- Locally available material.
- Application : Foundation



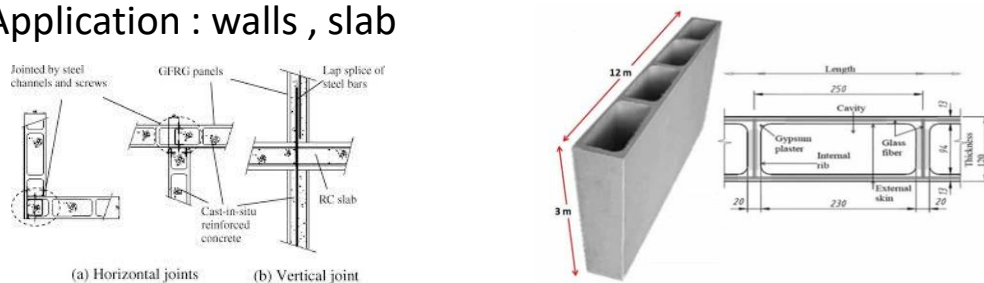
REINFORCED CEMENT CONCRETE (RCC)

- Application : Plinth beam
- SCREEN CONCRETE- FLOORING



GFRG PANELS

- Glass fiber reinforced gypsum (GFRG) walls are prefabricated large gypsum panels with hollow cores.
- Application : walls , slab



PUF PANELS

- These PUF Panels / PUF Sheets / Sandwich panels are manufactured in a computerised plant and come with Pre-Painted Galvanized Steel Sheets
- Application - roof



PAVER BLOCKS

- Application : pathways



ALTERNATIVE MATERIAL OSONA TREE – FLOORING MATERIAL



S.no	Materials	Quantity	Cost	Pictures
1.	Red Soil	2 bags	100rs Per bag	
2.	Cow Dunk	2 bags	Naturally availabl-e.	
3.	Osona Chali	5 kg	Naturally Availabl-e.	
4.	Cement	¼ of one bag	300 for one	



TENTATIVE COST OF IMPLEMENTATION

1	NAME	MATERIAL	UNITS	Per SQft/SQmt	COST
2	FOUNDATION	CONCRET	1500 cu.m	500/cu.m	747000
3		PCC	324cu.m	5000/cu.m	1620000
4		LATRITE BLOCK	46170 UNIT	30 per unitblock	1385100
5	PLINTH	RCC	20 cu.m		
6	WALL	GFRG PANEL			
7	DOOR	TEAK WOOD AND PVC	4.5 per cluster		
8	WINDOW	UPVC	4.5 per cluster		
9	FLOORING	SCREED CONCRET	11000sqmt		
10	ELECTRICAL			1350/sqft	174960000
11	PLUMBING				
12	SENITARY FITTING	CONCRET			
13	CHAJJA	GFRG PANEL			
14	KITCHEN PLATFORM	GRANITE AND CONCRET	369 sqft		
15	TOILET SEAT	CERAMIC	405		
16	SLAB	GFRG PANEL	12095sqmt		
17	PARAPET WALL	GFRG PANEL			
18	STAIRCASE	GFRG PANELS , CONCRET & BRIC	4050sqft	550/sqft	2227500
19	ROOF	PUF PANELS	2835sqmt	1400/sqmt	3969000
20	TOTAL AMOUNT FOR CLUSTER				184908600
21	TOTAL AMOUNT FOR 1 UNIT				456000
22	COST FOR SITE DEVELOPMENT		5% OF COST		9250000
23	ROAD	PERMEABLE PAVING , PREVIOUS CONCRET & POROUS ASPHALT	4320SQM		4320000
24	BIOGAS INFRASTRUCTURE	TANK , MACHINE, PIPELINE, GAS PIPELINE	20LAKHS FOR STRUCTURE	150% FOR PIPELINES AND MACHINE	5000000
25	SOLAR			RESCO MODEL	0
26	TOTAL			203934600	20.5 CRORE

BIO GAS PLANT

- **1kg** of human waste / organic waste can produce **0.4cub mt.** Of Biogas .
- **5kg** (approx)/day waste (includes all types of wastes) is generated from a family of **4-5 members**.
- **5 kg** of waste can produce **2cub.mt** (2000L) of biogas .
- **2cub.mt** gas can be used for 2 days of cooking for a family of 4-5 members .
- **1 cub.mt** biogas can produce 4kwh of electricity (50% is lost as heat , hence we get **2kwh**) .
- For a house of 30sqm , **5 bulbs 10watt, 2 fans 150watt** (in general) = **350watt (say 400watt)** .
- **2kwh** (2000wh) / **400watt** = **5hrs** of usage .

BIO GAS PLANT SIZE

- General requirement for a family of **4-5** members is **3cub.mt** of biogas.
- For production of **3 cub.mt** biogas volume **9 cub.mt** of digester volume is required .
- With 9cub.mt of biogas module 2.6m diameter cylindrical shape is required (height = 1.5m)
- **10cub.mt** biogas module cost **10,000/-**
- Biogas electric generator (3kva) cost **35,000/-**

CONCLUSION AND WAY FORWARD

The design is followed by 3 main aspects :

- Disaster resilient architecture

Understanding the needs of the user and keeping in mind the disaster resilient feature , we have taken inspiration from the ancient town planning for the form and orientation of the building blocks .

- Sustainability :

Trying to gain for a self dependent neighbourhood unit by applying new technologies required for smart city policies into the site.

1. Energy efficient : Solar street panels

2. Water efficient : waste water management (osho garden)

rain water harvesting by introducing Porous road – road which collects water and drain it to harvesting pit .

3. Solid waste management : by introducing BIO GAS PLANT .

- Barrier free design :

The term **barrier-free design** was first used in the 1950's to describe the effort of removing physical **barriers** from the "built environment" for people with disabilities.

Hence , ramps are provided where ever necessary , and park with features inducing barrier free design like tactile blocks, etc.



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LEARNINGS FROM THE HACKATHON

- 
- Accomodating a comfortable living space for marginalized people in a given space.
 - Gained an understanding of GOVERNMENT POLICIES on LIG , EWS , etc ., schemes for the better development of the smart cities.
 - A neighbourhood / community plan outlines a vision for the area and sets planning policies for the use and development of land.
 - Disaster risk reduction resilient architecture - building orientation , form and function of spaces , open spaces , etc.
 - Application of new technologies , researching and approx. calculations on BIO GAS PLANT , solar street lights.